Interval notation

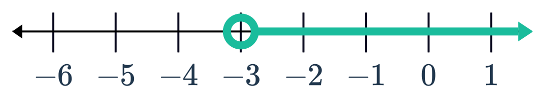
1 Write the interval notation for each of the following pictured inequalities:

a

b

c

d



e

2 For each of the following inequalities express the solutions of the inequality using interval notation:

a $x < -7$ **b** $x \leq 6$ **c** $x \geq -4$ **d** $-7 \leq x \leq 3$

Domain and range

3 For each of the following relations:

i Find the domain.

ii Find the range.

iii Determine whether the relation is a function or not.

a $\{(8,5),(1,3),(3,4),(9,1),(2,9)\}$

b $\{(2,3),(3,8),(6,1),(8,7),(3,2)\}$

c $\{(4,4),(8,9),(2,8),(7,9),(3,1)\}$

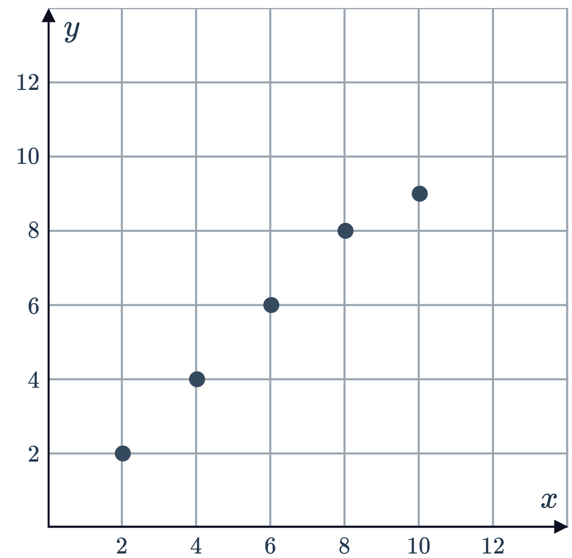
d

x	1	6	3	8	2
y	3	2	7	1	2

e

x	7	7	8	5	3
y	1	9	3	2	6

f



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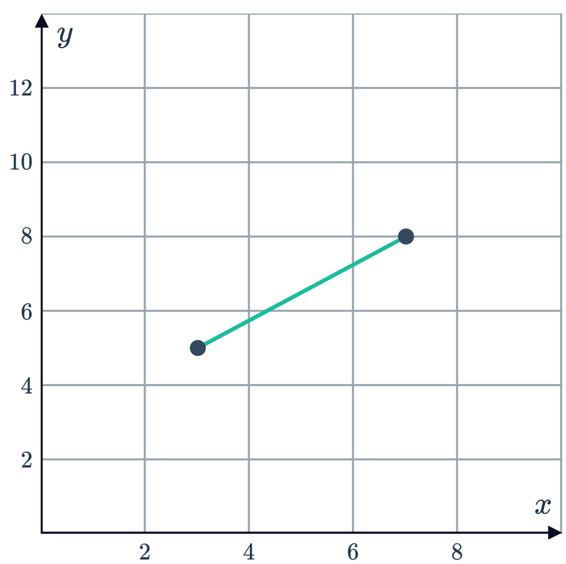

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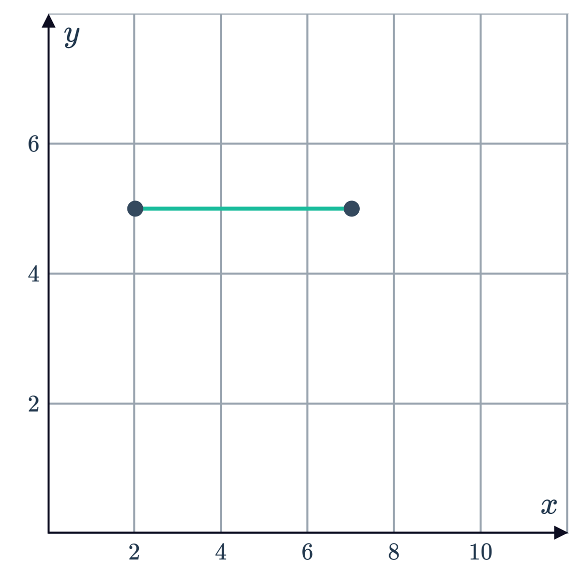
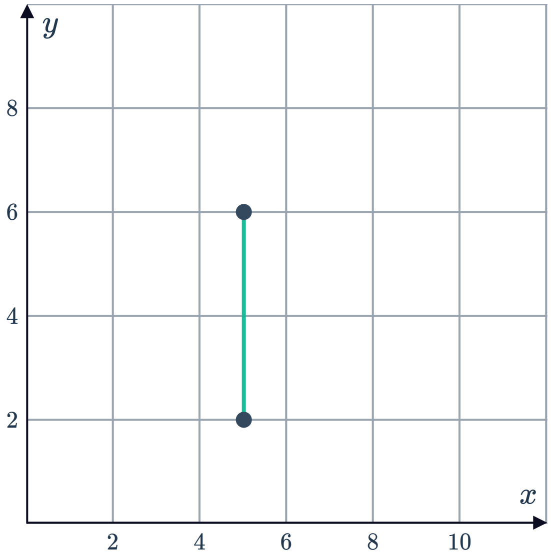
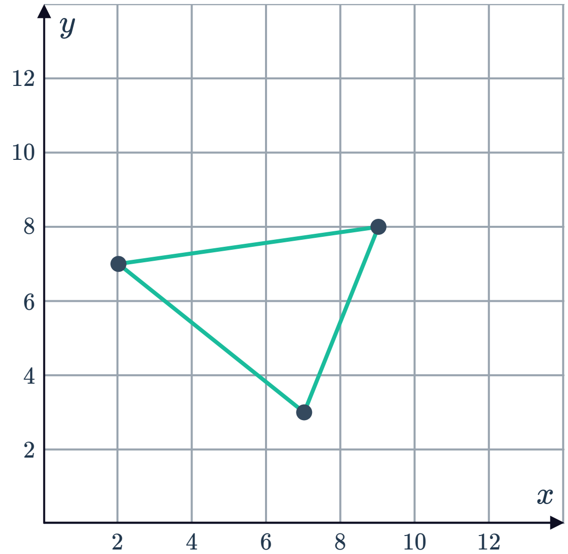


i





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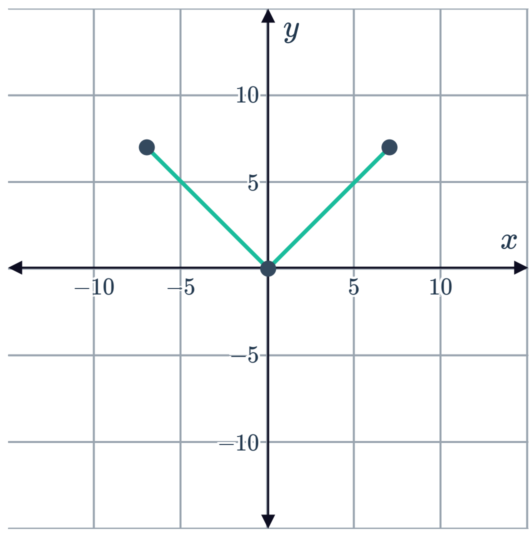

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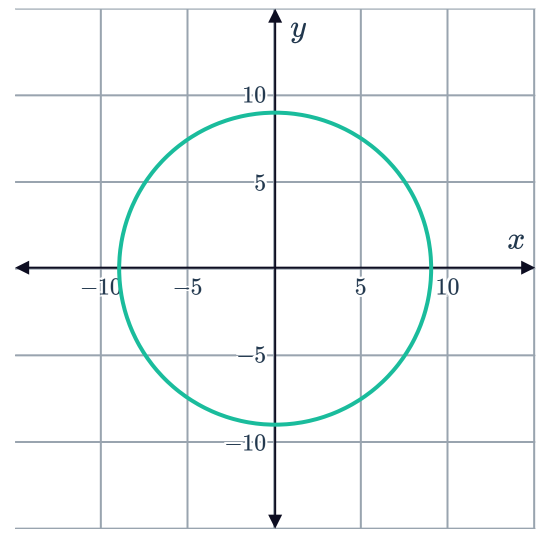

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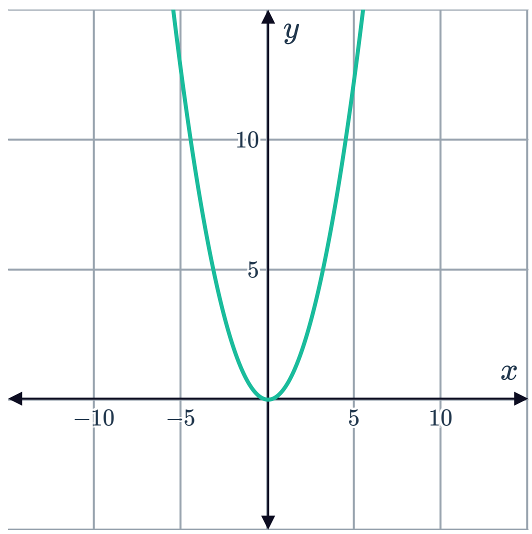
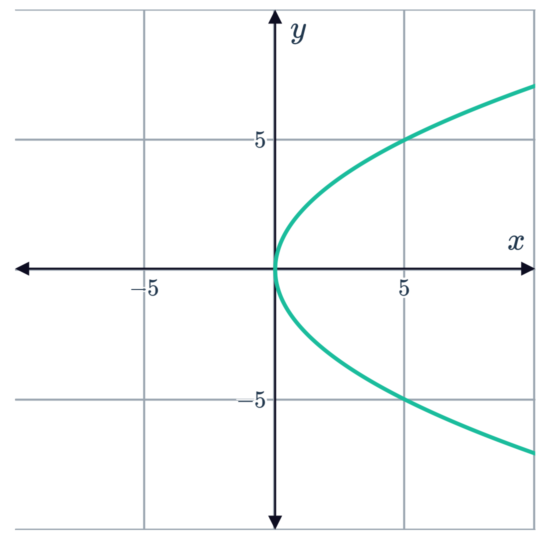
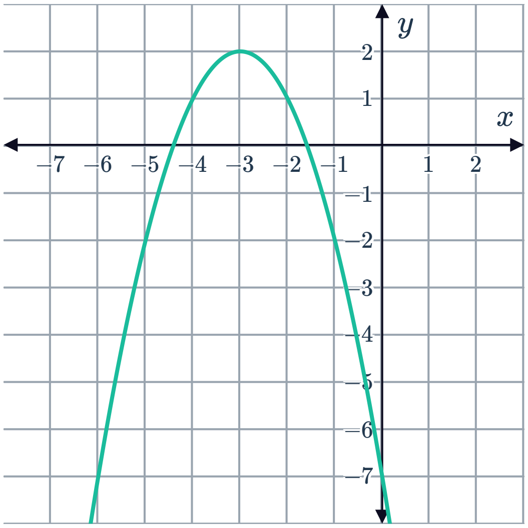


n

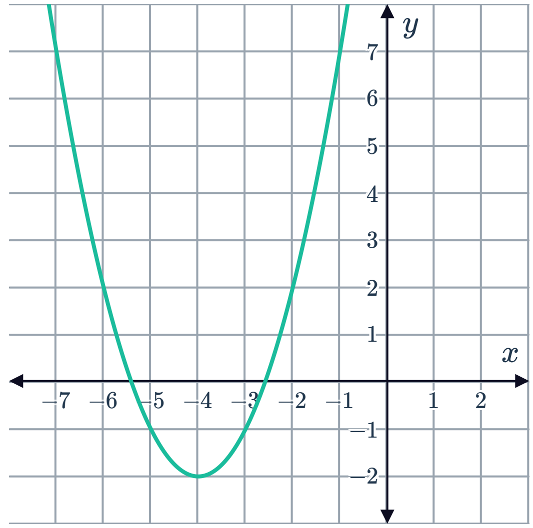


4 State the domain and range of the following[redacted]tions:

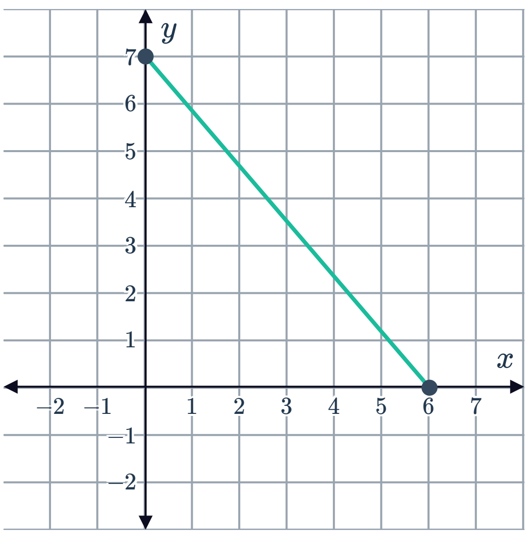
a



b



c

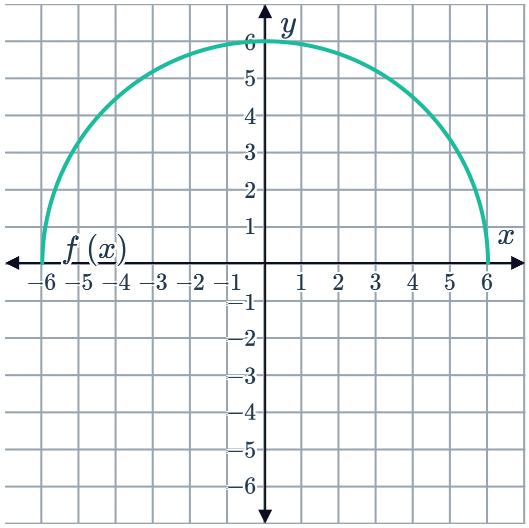


5 State the domain of the following functions:

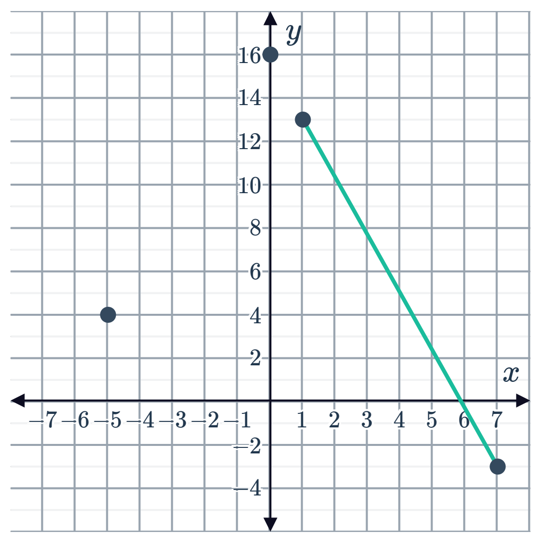
a $f(x) = -3x - 1$

b $f(x) = 8x + 8$

c



6 Consider the function f graphed. Use the graph to determine whether each of the given statements are true or false:



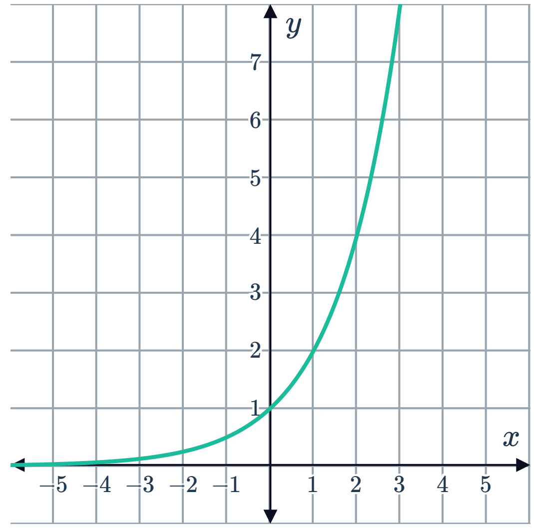
a The domain is $[-5, 7]$

b The range is $[-3, 16]$

c $f(1) - f(7) = 16$

d $f(0) = 16$

7 Consider the graph of $y = 2^x$:



a State the y-intercept of the graph.

b Does the graph have an x-intercept?

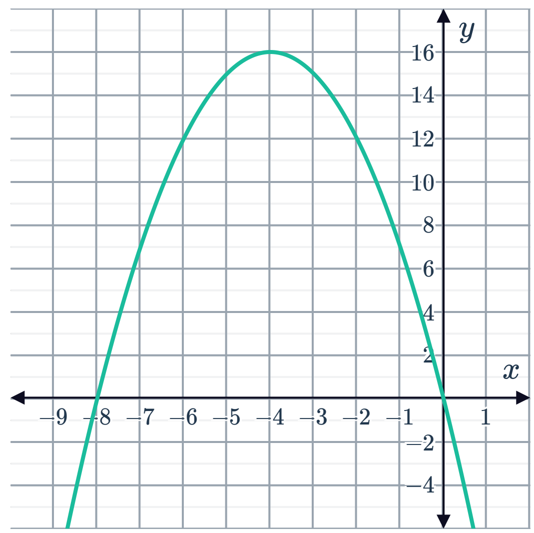
c State the domain of the function.

d State the range of the function.

e Find the value of y when $x = 7$.

f Find the value of x when $y = 256$.

8 Consider the graph of the function shown:

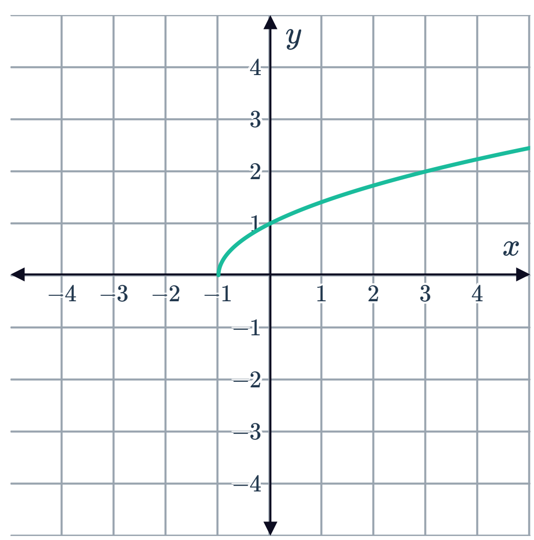


a State the maximum value of the graph.

b State the range of the function.

c State the domain of the function.

9 Consider the graph of the function $f(x) = \sqrt{x+1}$:



a State the domain of the function.

b Is there a value in the domain that can produce a function value of -2?

10 Consider the function $f(x) = (x-2)^2$, for a[redacted] x.

a Graph the function.

b Is there a value in the domain that can produce a function value of -9?

Continuity

11 A function $f(x)$ is known to be continuous on the interval $(-\infty, \infty)$.

a Is $f(x)$ continuous over all smaller intervals?

b Suppose the function is $f(x) = 3x + 8$. State whether $f(x)$ is continuous on the following intervals:

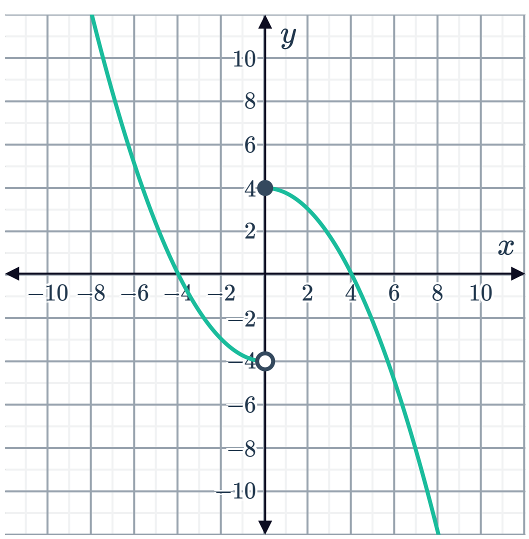
i $[4, 6]$ ii $(-7, -1)$ iii $(0, \infty)$

12 Consider the function $f(x) = 4x^3 - 9x^2 - 2x + 1$. State whether $f(x)$ is continuous over the following intervals:

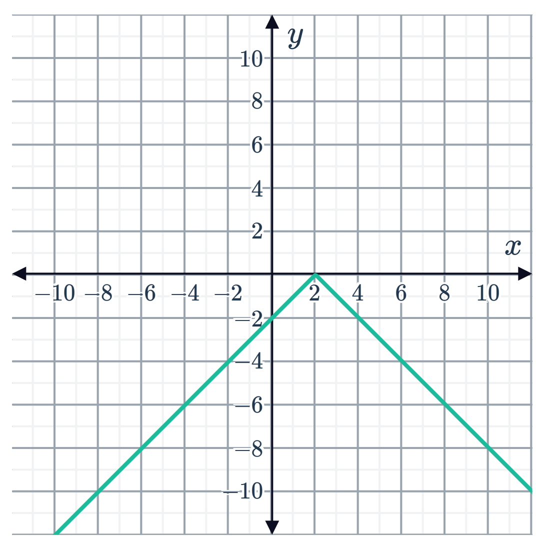
a $(-\infty, \infty)$ b $[9, 15]$

13 State whether the following functions are continuous:

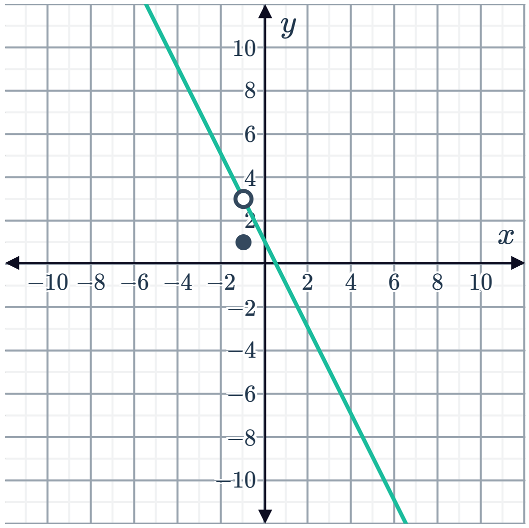
a

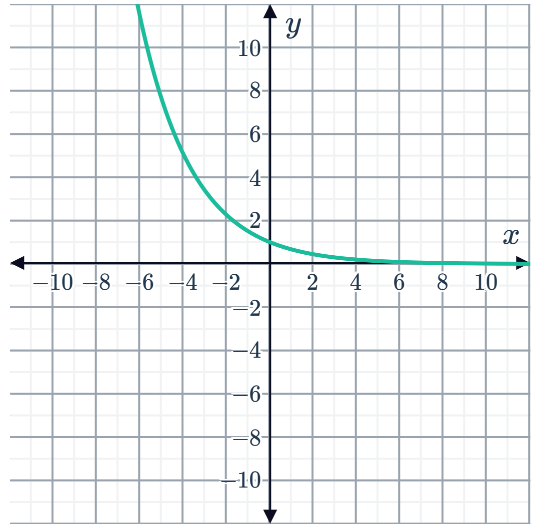


b

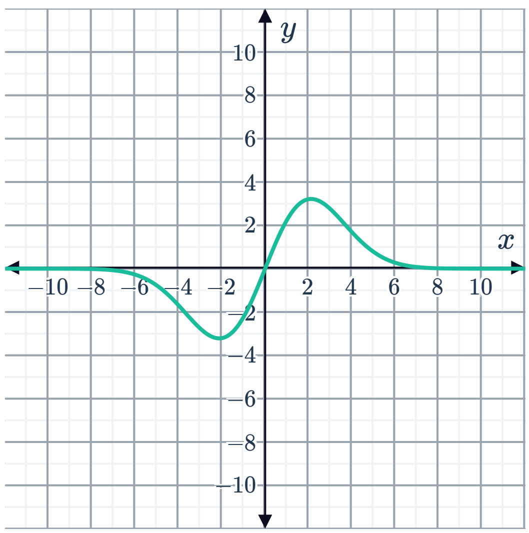


c

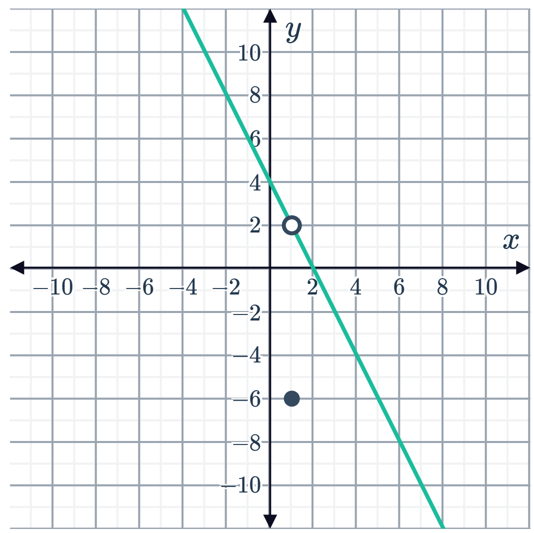

y
x
−10 −8 −6 −4 −2 2 4 6 8 10
10 8 6 4 2 −2 −4 −6 −8 −10


y
x
−10 −8 −6 −4 −2 2 4 6 8 10
10 8 6 4 2 −2 −4 −6 −8 −10

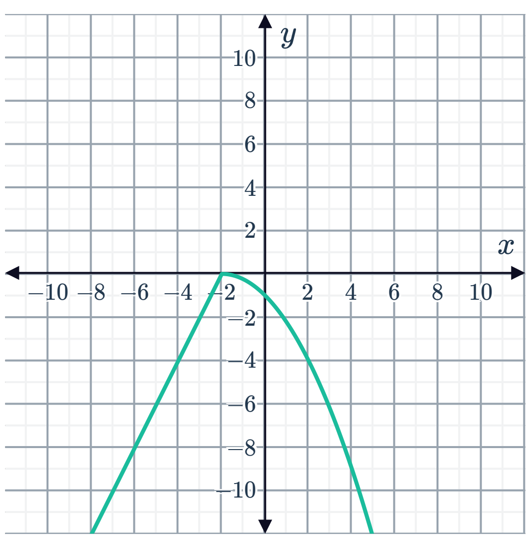
e


y
x
−10 −8 −6 −4 −2 2 4 6 8 10
10 8 6 4 2 −2 −4 −6 −8 −10

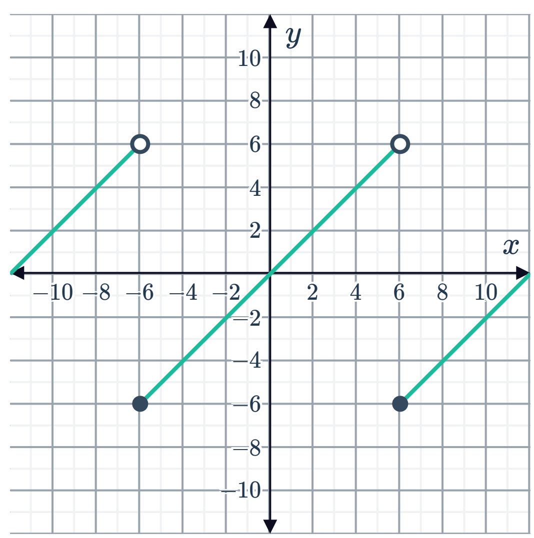
f


y
x
−10 −8 −6 −4 −2 2 4 6 8 10
10 8 6 4 2 −2 −4 −6 −8 −10

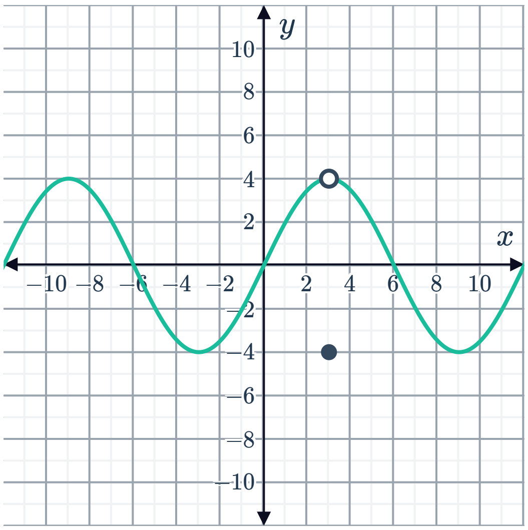
g

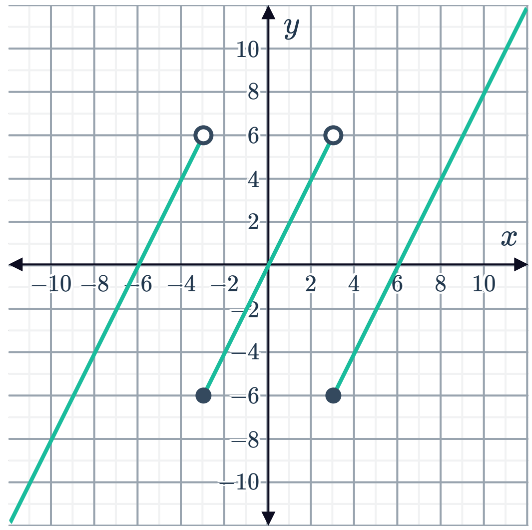

y
x
−10 −8 −6 −4 −2 2 4 6 8 10
10 8 6 4 2 −2 −4 −6 −8 −10

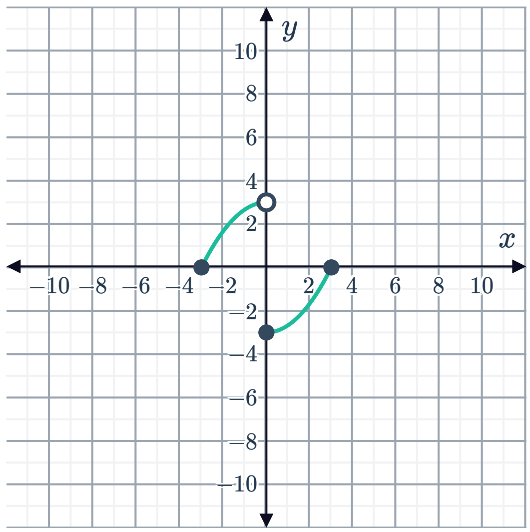
h


y
x
−10 −8 −6 −4 −2 2 4 6 8 10
10 8 6 4 2 −2 −4 −6 −8 −10

14 Consider the following graphs of functions:







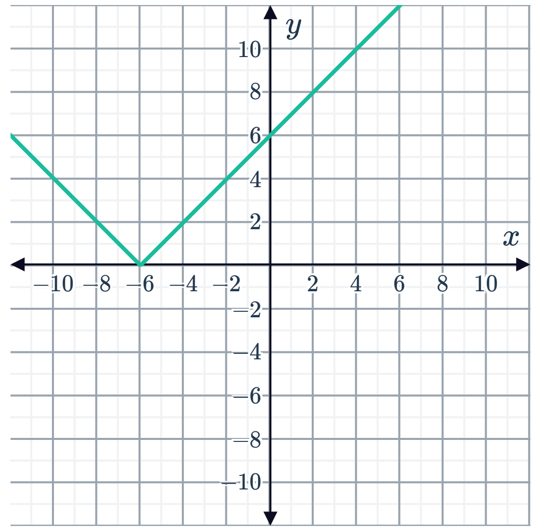
State whether the following statements are true or false:

a Some of the functions are continuous.

b All of the functions have at least one discontinuous point in their domain.

c All of the functions have a domain of $(-\infty, \infty)$.

15 Consider the graph of the function $f(x) =$ [illegible]:
State whether the function is continuous over the following intervals:

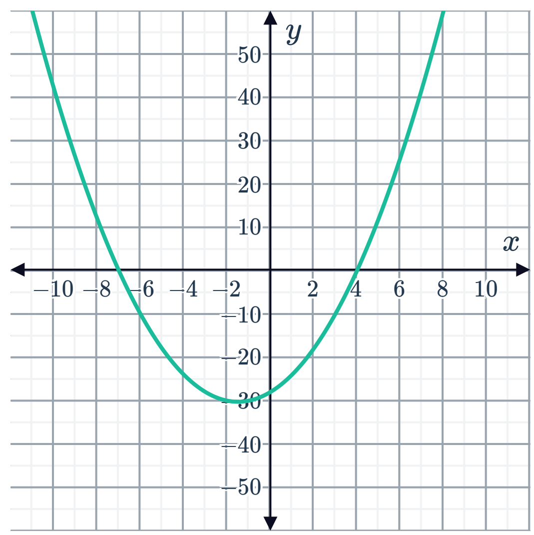


a $(-6, \infty)$

b $(-\infty, -6)$

c $(-\infty, \infty)$

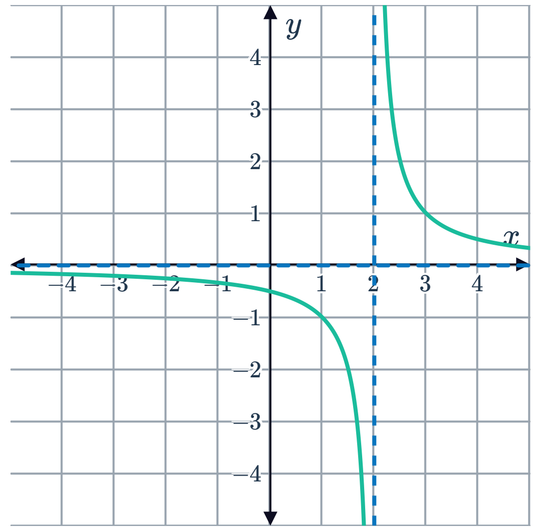
16 Consider the graph of the function $f(x) = (x + 7)(x - 4)$:
State whether $f(x)$ is a continuous function over:



a The domain.

b The interval $[-7, 4]$

17 Consider the graph of the function $f(x) = \frac{\text{[redacted]}}{x-2}$:

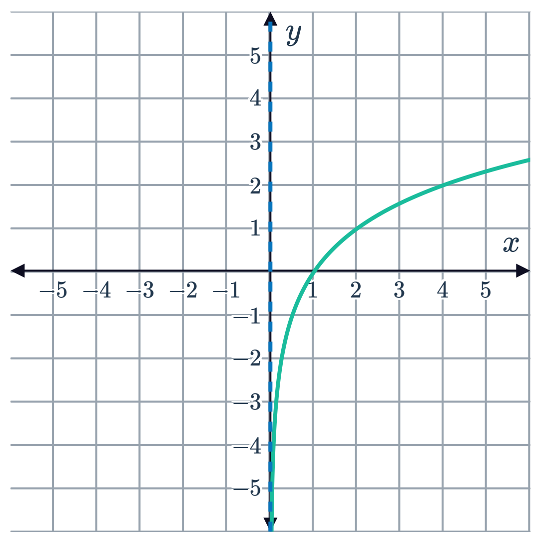


a State the domain of $f(x)$.

b Is $f(x)$ continuous over the interval $(-\infty, 2) \cup (2, \infty)$?

c Is $f(x) = \dfrac{1}{x-2}$ a continuous function over its domain?

18 Consider the graph of the function $f(x) = \log_2 x$:

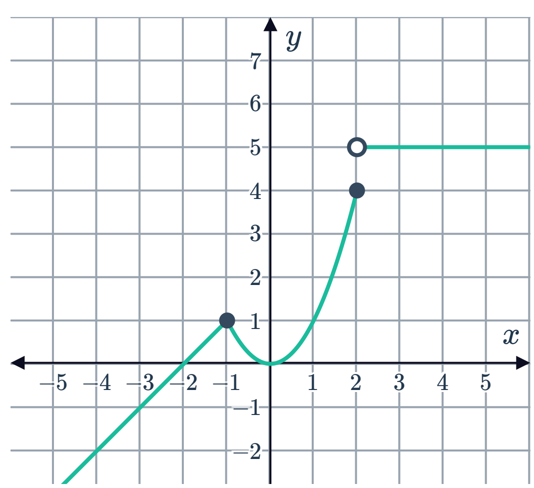


a State the domain of $f(x)$.

b Is $f(x)$ continuous over the interval $(0, \infty)$?

c Is $f(x) = \log_2 x$ a continuous function over its domain?

19 Consider the function $f(x)$ shown:



a Is $f(x)$ continuous over its domain $(-\infty, \infty)$?

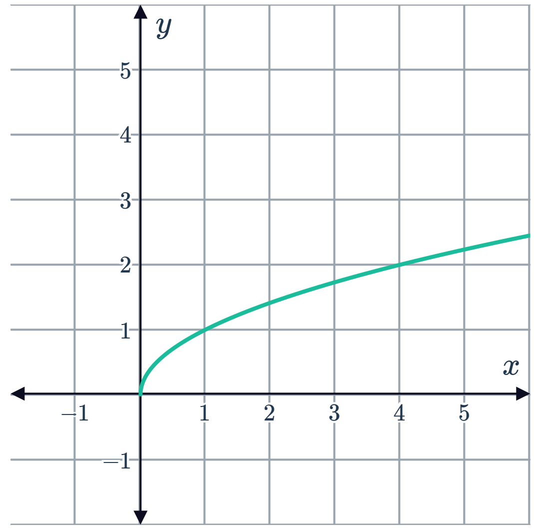
b State the largest interval over which $f(x)$ is continuous.

20 Consider the graph of the function $f(x) = \sqrt{x}$:

a State the domain of $f(x)$.

b State the largest interval over which $f(x)$ is continuous.

c Is $f(x) = \sqrt{x}$ a continuous function over its domain?

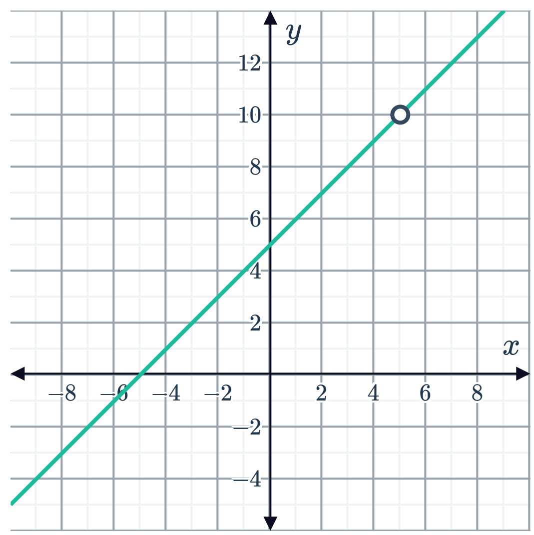


21 Consider the graph of the function $f(x) = \dfrac{x^2 - 25}{x - 5}$:

a State the domain of $f(x)$.

b State the largest interval over which $f(x)$ is continuous.

c Is $f(x) = \dfrac{x^2 - 25}{x - 5}$ a continuous function over its domain?

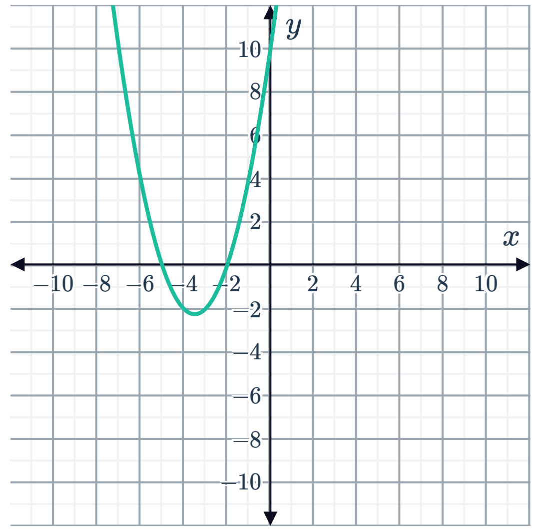


22 Consider the graph of the function $f(x) =$ [redacted] $x + 10$:

a State the domain of $f(x)$.

b State the largest interval over which $f(x)$ is continuous.

c Is $f(x) = x^2 + 7x + 10$ a continuous function over its domain?

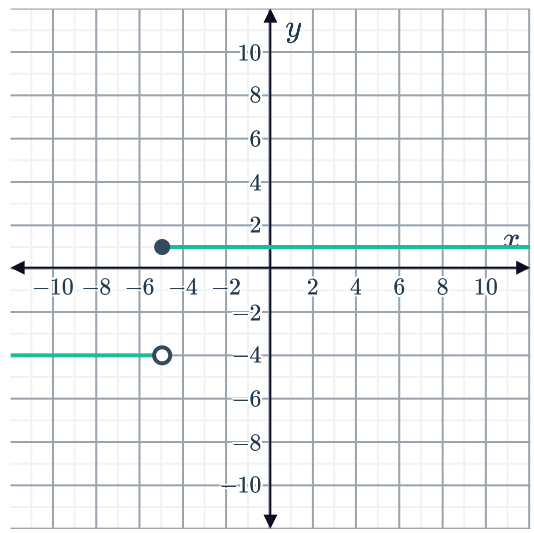


23 Consider the step function $f(x)$ shown:

a State the domain of $f(x)$.

b State the largest interval over which $f(x)$ is continuous.

c Is $f(x)$ a continuous function over its domain? Explain your answer.

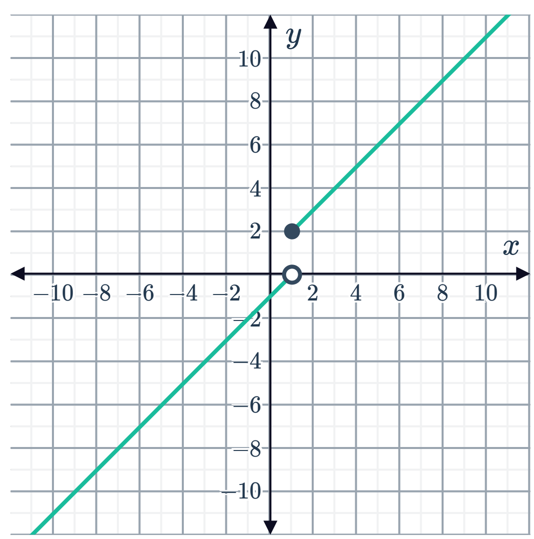


24 Consider the function $f(x)$ shown:

a State the domain of $f(x)$.

b State the largest interval over which $f(x)$ is continuous.

c Is $f(x)$ a continuous function over its domain? Explain your answer.



Applications

25 The function $f(n)$ is used to determine the area of a square given its side length.

a State whether the following values are part of the domain of the function:

i -8 ii 6.5 iii $9\frac{1}{3}$ iv $\sqrt{78}$

b For $n \geq 0$, state the area function for a side length of n.

c Sketch the graph of the function f.

26 The number of bees in a colony is initially measured and left to form a hive. The number of bees in the hive is measured each day over the course of one week. The function

$$H(x) = 30(3)^x$$

is found to model the number of bees, H, after x days.

a What is the domain of the function?

b What is the initial number of bees in the colony?

c Can the function be used to model the number of bees over an entire season?

27 The water level in a dam is measured to be 190 yd. After two weeks, the water level is measured again and found to be 140 yd.

a Is there a time during the two weeks that the water level of the dam is at 160 yd?

b To answer part (a) what property do we assume about the water level over the two weeks?

c Consider a function $f(x)$ with domain $[0,2]$, where $f(0) = 190$ and $f(2) = 140$. If the function is known to be discontinuous, does the function take a value of 160 at some point in its domain? Explain your answer.